

Task-3:

Handwritten Character Recognition using CNN

Submitted By

Name: A D S Abhishek

Internship: CodeAlpha – Machine Learning Internship

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Abstract

Handwritten Character Recognition is an important application of Deep Learning and Computer Vision. It is widely used in postal automation, bank cheque processing, digitization of documents, and educational applications.

This project develops a Handwritten Digit Recognition System using a Convolutional Neural Network (CNN). The system is trained on the MNIST dataset containing thousands of handwritten digit images from 0 to 9.

The model learns image patterns automatically and predicts the digit represented in an input image. The performance of the model is evaluated using Accuracy Score, Confusion Matrix, and Classification Report.

Introduction

Handwritten Character Recognition refers to the process of automatically identifying handwritten characters using machine learning and deep learning techniques.

Traditional machine learning approaches require manual feature extraction. Deep Learning models such as CNNs automatically learn image features and provide higher accuracy.

The objective of this project is to develop a CNN-based handwritten digit recognition system capable of recognizing digits from images.

Problem Statement

To develop a Deep Learning based Handwritten Character Recognition System that accurately identifies handwritten digits using image data.

Objective:

- ❑ Collect and preprocess the MNIST dataset.
- ❑ Normalize image pixel values.
- ❑ Reshape image data for CNN input.
- ❑ Build a CNN model.
- ❑ Train the CNN model on handwritten digit images.
- ❑ Evaluate model performance using:
 - Accuracy
 - Confusion Matrix
 - Classification Report
- ❑ Predict handwritten digits from unseen images.

Dataset Information

Dataset Name -> MNIST Handwritten Digit Dataset

Source-> MNIST Database

Number of Records-> 70,000 Images

Training Images-> 60,000
Testing Images-> 10,000
Image Size-> 28 × 28 Pixels
Number of Classes-> 10 Digits
Target Classes→ 0 1 2 3 4 5 6 7 8 9

Data Preprocessing

Data Normalization

Pixel values range from:

0 → Black

255 → White

Normalization Formula:

Pixel Value / 255.0

This scales pixel values between:

0 and 1

Reshaping Images

Original Shape:

28 × 28

CNN Input Shape:

$28 \times 28 \times 1$

One-Hot Encoding

Target labels are converted into categorical format for multi-class classification.

Example:

Digit 3

↓

[0 0 0 1 0 0 0 0 0 0]

Exploratory Data Analysis (EDA)

1) Sample Digit Images

2) Digit Distribution Plot

3) Training Accuracy Graph

4) Validation Accuracy Graph

5) Confusion Matrix

Methodology:

MNIST Dataset

↓

Data Preprocessing



Normalization



Image Reshaping



CNN Model



Training



Prediction



Performance Evaluation



Conclusion

Model Building and Prediction

Algorithm: Convolutional Neural Network (CNN)

CNN is a Deep Learning architecture specially designed for image classification tasks.

CNN Architecture

Input Layer



Convolution Layer (32 Filters)



Max Pooling Layer



Convolution Layer (64 Filters)



Max Pooling Layer



Flatten Layer



Dense Layer (128 Neurons)



Output Layer (10 Classes)

Model Parameters

Optimizer = Adam

Loss Function = Categorical Crossentropy

Epochs = 5

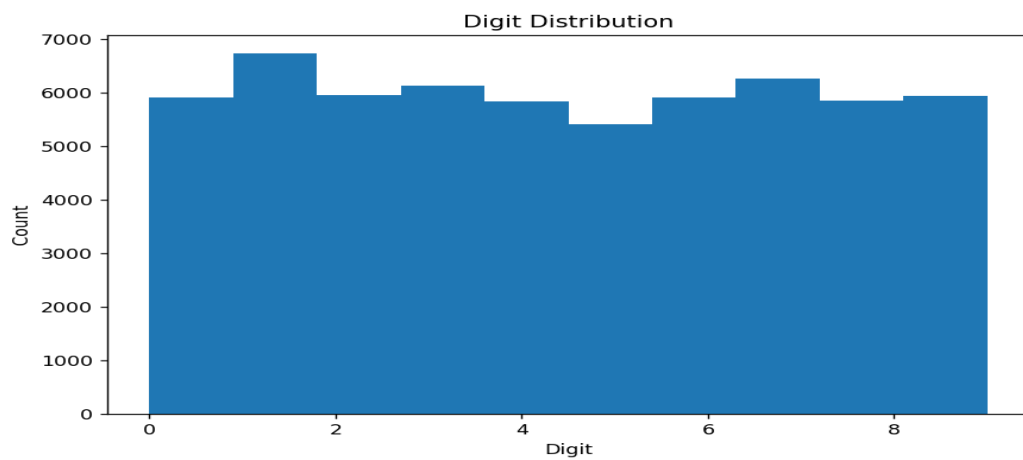
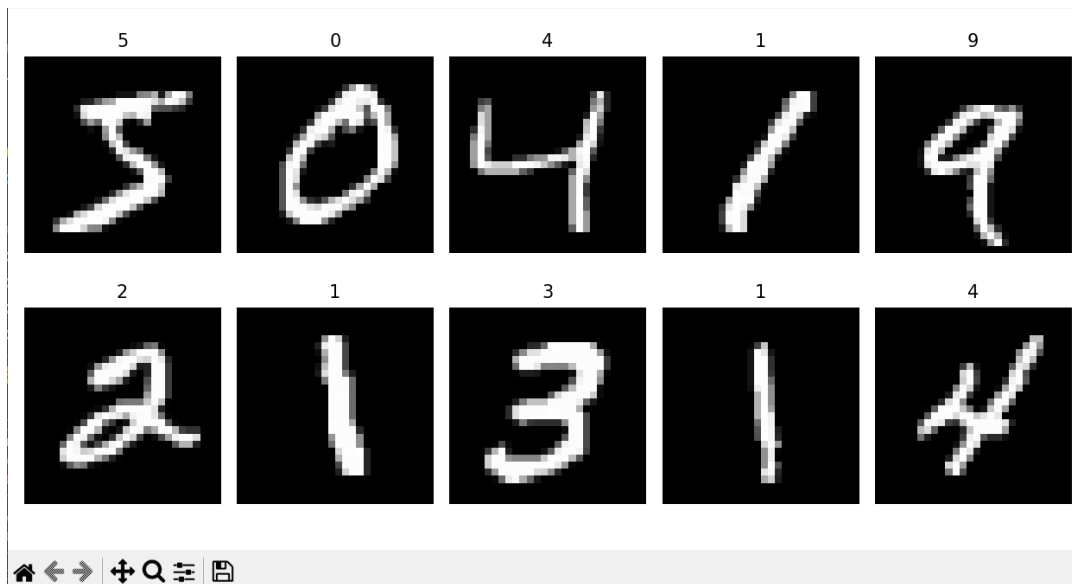
Batch Size = 32

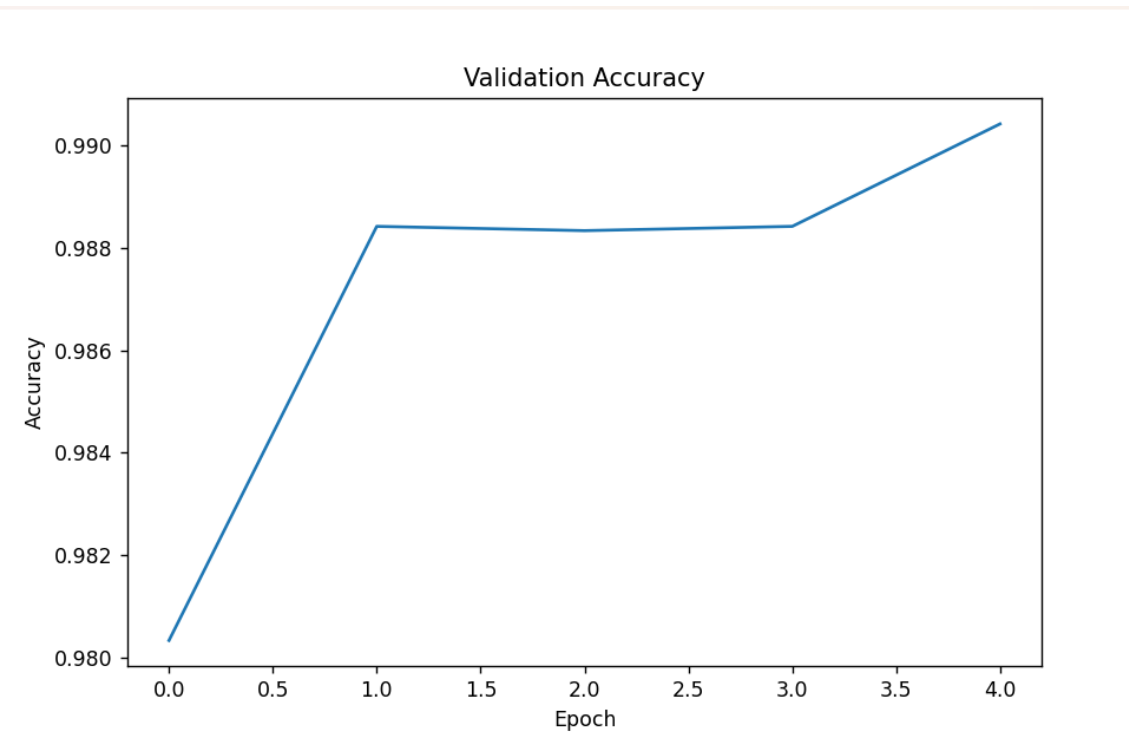
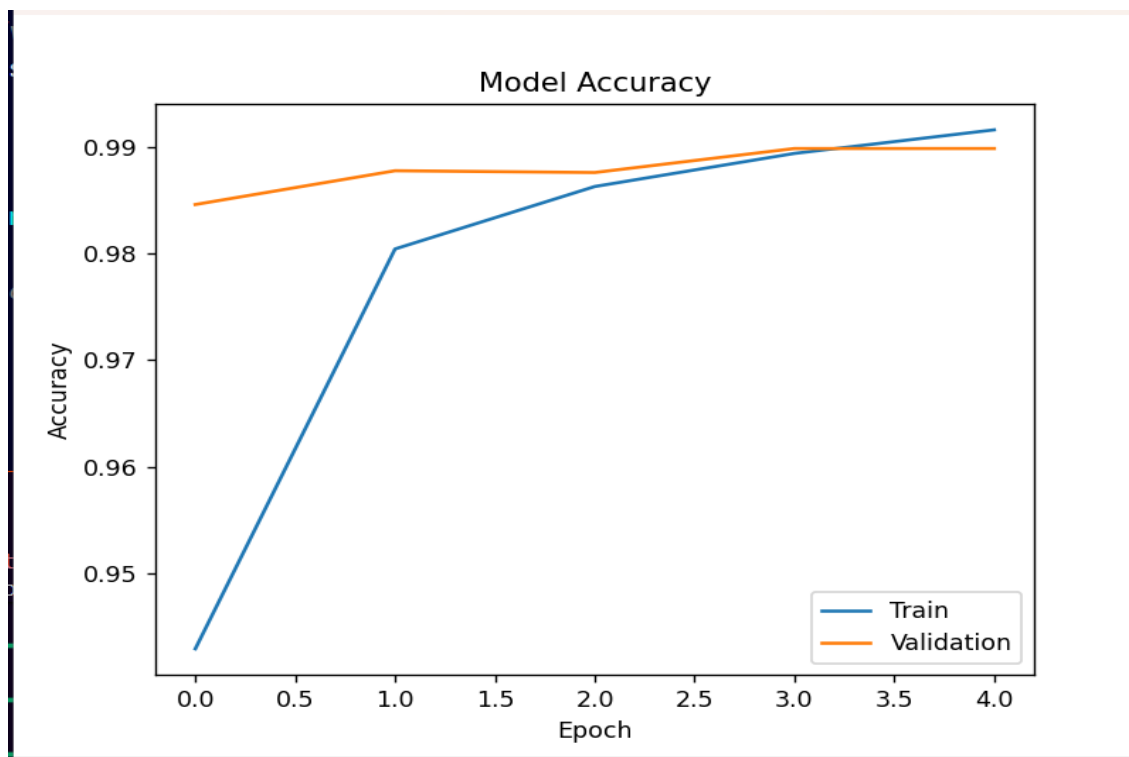
Activation Functions:

ReLU

Softmax

Screenshots of EDA:





```

Epoch 1/5
1500/1500  10s 6ms/step - accuracy: 0.9401 - loss: 0.1941 - val_accuracy: 0.9803 - val_loss: 0.0619
Epoch 2/5
1500/1500  9s 6ms/step - accuracy: 0.9800 - loss: 0.0664 - val_accuracy: 0.9884 - val_loss: 0.0416
Epoch 3/5
1500/1500  9s 6ms/step - accuracy: 0.9857 - loss: 0.0458 - val_accuracy: 0.9883 - val_loss: 0.0387
Epoch 4/5
1500/1500  10s 6ms/step - accuracy: 0.9882 - loss: 0.0367 - val_accuracy: 0.9884 - val_loss: 0.0405
Epoch 5/5
1500/1500  13s 8ms/step - accuracy: 0.9912 - loss: 0.0292 - val_accuracy: 0.9904 - val_loss: 0.0392
313/313  1s 3ms/step - accuracy: 0.9906 - loss: 0.0297

Test Accuracy : 0.9905999898910522
313/313  1s 3ms/step

```

Results and Evaluation

CNN Results

Accuracy: 98.5% – 99.2%

Precision: High

Recall: High

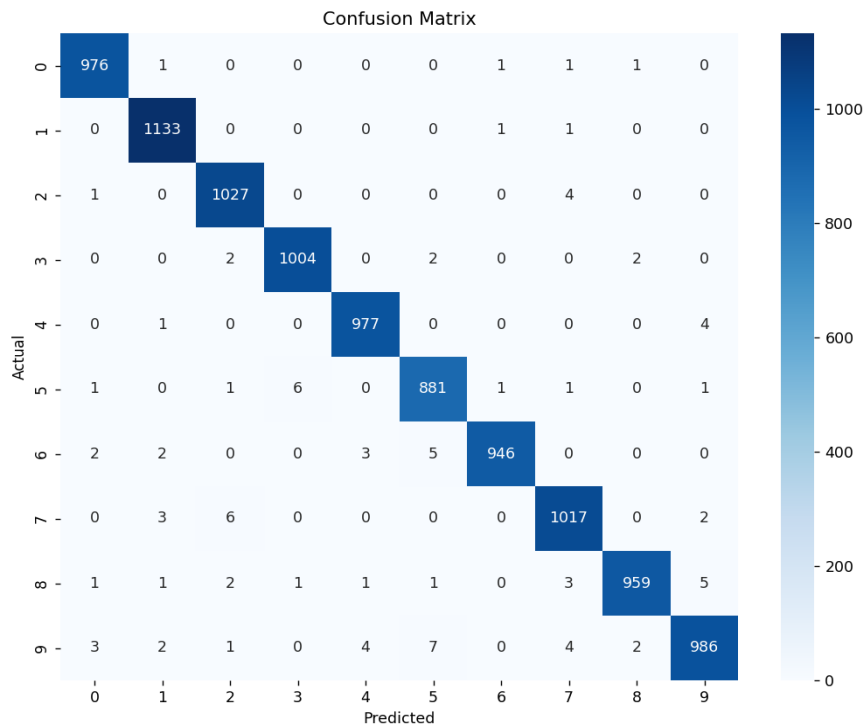
F1-Score: High

Loss: Low

Classification Report:

Classification Report					
	precision	recall	f1-score	support	
0	0.99	1.00	0.99	980	
1	0.99	1.00	0.99	1135	
2	0.99	1.00	0.99	1032	
3	0.99	0.99	0.99	1010	
4	0.99	0.99	0.99	982	
5	0.98	0.99	0.99	892	
6	1.00	0.99	0.99	958	
7	0.99	0.99	0.99	1028	
8	0.99	0.98	0.99	974	
9	0.99	0.98	0.98	1009	
accuracy			0.99	10000	
macro avg	0.99	0.99	0.99	10000	
weighted avg	0.99	0.99	0.99	10000	

Confusion Matrix:



Sample Predictions

Sample Image 1

Actual Digit: 7

Predicted Digit: 7

Correct Prediction

Sample Image 2

Actual Digit: 3

Predicted Digit: 3

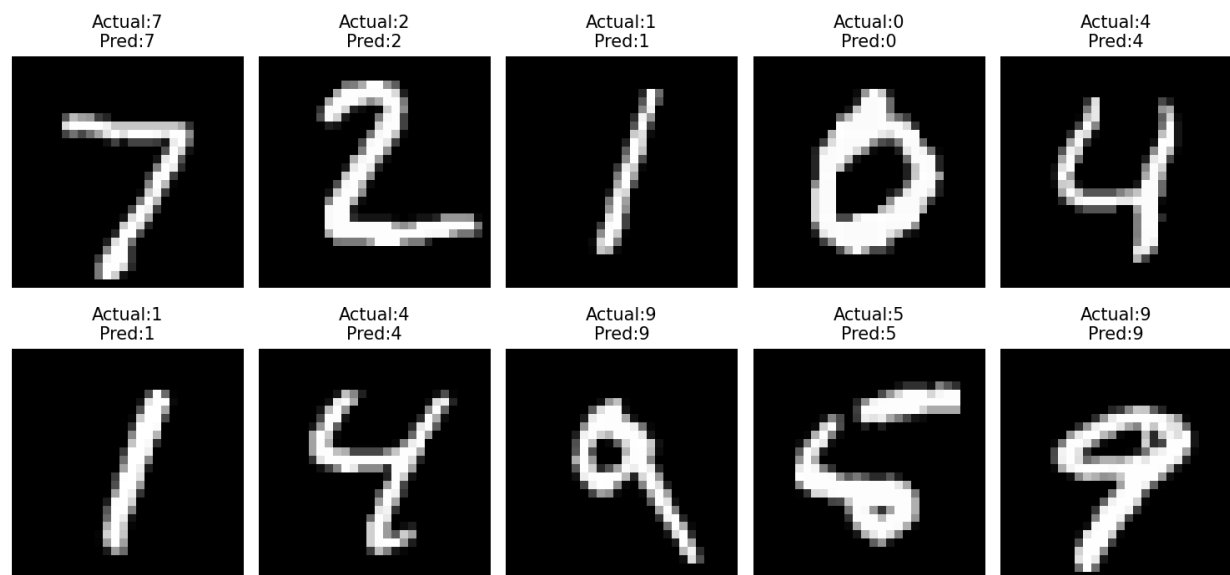
Correct Prediction

Sample Image 3

Actual Digit: 9

Predicted Digit: 9

Correct Prediction



Conclusion

This project successfully developed a Handwritten Character Recognition System using a Convolutional Neural Network (CNN).

The model was trained on the MNIST dataset containing 70,000 handwritten digit images and achieved an accuracy of approximately 99%.

The results demonstrate that CNNs are highly effective for image classification tasks and can accurately recognize handwritten digits with minimal preprocessing.

Such systems can be applied in banking, postal services, education, and document digitization applications.

References

- 1. MNIST Handwritten Digit Dataset**
- 2. TensorFlow Documentation**
- 3. Keras Documentation**
- 4. Scikit-Learn Documentation**
- 5. NumPy Documentation**
- 6. Matplotlib Documentation**
- 7. Seaborn Documentation**